Assignment 3

Question 1. Deutsch-Jozsa Algorithm for $n = 2$ (20 points)

The Deutsch-Jozsa Algorithm determines whether a function $f:\{0,1\}^n \to \{0,1\}$ is constant (same output for all inputs) or balanced (half outputs are 0, half are 1) using a single oracle query. In this question we work out the case $n = 2$, with $2^n = 4$ possible inputs $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

Suppose that the initial 2-qubit input state is initialized in $|00\rangle$ and the ancilla is initialized to $|1\rangle$ giving total initial state $|\psi_1\rangle = |00\rangle \otimes |1\rangle = |001\rangle$.

a. **(4 points)** Complete the following table of all functions of the form $f:\{0,1\}^2 = \{00,01,10,11\} \to \{0,1\}$ that are either constant or balanced.

Function	$f(0,0)$	$f(0,1)$	$f(1,0)$	$f(1,1)$	Constant/Balanced
f_1	0	0	0	0	Constant
f_2					Constant
f_3	0	0	1	1	Balanced
f_4					Balanced
f_5	0	1	1	0	Balanced
f_6					Balanced
f_7					Balanced
f_8	1	1	0	0	Balanced

b. **(4 points)** Applying $\left(\hat{H}^{\otimes 2} \otimes \hat{H}\right) = \hat{H} \otimes \hat{H} \otimes \hat{H}$ on $|\psi_1\rangle$ gives

$$|\psi_2\rangle = \left(\hat{H} \otimes \hat{H} \otimes \hat{H}\right)|001\rangle = \hat{H}|0\rangle \otimes \hat{H}|0\rangle \otimes \hat{H}|1\rangle = |+\rangle \otimes |+\rangle \otimes |-\rangle.$$

Write the 2-qubit input state in computational basis $|0\rangle, |1\rangle$ to show that $|\psi_2\rangle$ is of the form

$$|\psi_2\rangle = \frac{1}{2} \sum_{x \in \{0,1\}^2} |x\rangle \otimes |-\rangle = \frac{1}{2} \sum_{\substack{x_1 \in \{0,1\} \\ x_2 \in \{0,1\}}} |x_1\rangle \otimes |x_2\rangle \otimes |-\rangle.$$

c. **(2 points)** In the Deutsch Algorithm, we saw that the oracle operator $\hat{U}_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$ when acting on $|x\rangle$ in computational basis states and ancilla in state $|y\rangle = |-\rangle$ led to $\hat{U}_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$. For general 2-qubit state, this generalizes to

$$|\psi_3\rangle = \hat{U}_f|\psi_2\rangle = \frac{1}{2} \sum_{x \in \{0,1\}^2} (-1)^{f(x)}|x\rangle \otimes |-\rangle$$

$$= \frac{1}{2} \sum_{\substack{x_1 \in \{0,1\} \\ x_2 \in \{0,1\}}} (-1)^{f(x_1,x_2)}|x_1\rangle \otimes |x_2\rangle \otimes |-\rangle.$$

Can you justify this generalization without calculation?

d. **(2 points)** For constant $f(x_1, x_2) = c$, $c \in \{0,1\}$,

$$|\psi_3\rangle = \frac{1}{2}(-1)^c \sum_{\substack{x_1 \in \{0,1\} \\ x_2 \in \{0,1\}}} |x_1\rangle \otimes |x_2\rangle \otimes |-\rangle = (-1)^c|+\rangle \otimes |+\rangle \otimes |-\rangle.$$

Can you justify the last step without a new calculation? Hint: Revisit part (b).

e. **(2 points)** For constant $f(x_1, x_2) = c$ case, apply $\hat{H} \otimes \hat{H} \otimes \hat{I}$ to get $|\psi_4\rangle$ and calculate probability of measuring $|00\rangle$ for the first two qubits.

f. **(6 points)** As an example of a balanced function, consider f_3:

$$f_3(0,0) = f_3(0,1) = 0, \qquad f_3(1,0) = f_3(1,1) = 1.$$

For f_3 expand the following in computational basis

$$|\psi_3\rangle = \frac{1}{2} \sum_{\substack{x_1 \in \{0,1\} \\ x_2 \in \{0,1\}}} (-1)^{f_3(x_1,x_2)} |x_1\rangle |x_2\rangle |-\rangle.$$

Then apply $\left(\hat{H} \otimes \hat{H} \otimes \hat{I}\right)$ to obtain $|\psi_4\rangle$. Show that in this case probability of measuring $|00\rangle$ for the first two qubits is 0. Hint: You only need one Walsh-Hadamard Transform for this problem.

You may find the following useful:

1. Hadamard Gate: $\hat{H}|0\rangle = |+\rangle,\ \hat{H}|1\rangle = |-\rangle,\ \hat{H}|+\rangle = |0\rangle,\ \hat{H}|-\rangle = |1\rangle$
2. Computational Basis (orthonormal) for 2-Qubit Systems: $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$
3. Walsh-Hadamard Transform for 2 Qubits

$$\left(\hat{H} \otimes \hat{H}\right)|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$
$$\left(\hat{H} \otimes \hat{H}\right)|01\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$
$$\left(\hat{H} \otimes \hat{H}\right)|10\rangle = \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle)$$
$$\left(\hat{H} \otimes \hat{H}\right)|11\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$$

Since $\hat{H}^2 = \hat{I}$, applying $\left(\hat{H} \otimes \hat{H}\right)$ inverts the transform.

Question 2. Density Matrices (12 points)

Consider the pure state $|\phi\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\ i\end{bmatrix}$.

a. **(1 points)** Write density matrix for state $|\phi\rangle$ in matrix representation.

b. **(1 points)** Prove that this density matrix is normalized, i.e., $\mathrm{Tr}[\hat{\rho}] = 1$.

c. **(2 points)** Prove that this density matrix is Hermitian, i.e., $\hat{\rho}^\dagger = \hat{\rho}$.

d. **(2 points)** Show that this density matrix represents a pure state, i.e., $\mathrm{Tr}[\hat{\rho}^2] = 1$.

Consider mixed state $\hat{\rho} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$.

e. **(1 points)** Write this density matrix in matrix representation.

f. **(1 points)** Prove that this density matrix is normalized, i.e., $\mathrm{Tr}[\hat{\rho}] = 1$.

g. **(2 points)** Prove that this density matrix is Hermitian, i.e., $\hat{\rho}^\dagger = \hat{\rho}$.

h. **(2 points)** Show that this density matrix represents a mixed state, i.e., $\mathrm{Tr}[\hat{\rho}^2] < 1$.

Question 3. Qubit-Qubit Interaction (18 points)

Consider two qubits $|\psi_1\rangle = |0\rangle$ and $|\psi_2\rangle = |0\rangle$, so that their combined state is

$$|\psi_0\rangle = |\psi_1\rangle \otimes |\psi_2\rangle = |0\rangle \otimes |0\rangle = \frac{|++\rangle + |+-\rangle + |-+\rangle + |--\rangle}{2}.$$

Suppose they interact via the following interaction Hamiltonian

$$\widehat{H}_{int} = \hbar g \hat{X} \otimes \hat{X}.$$

Then the time evolution operator is

$$\widehat{U}(t) = e^{-i\widehat{H}_{int}t/\hbar} = e^{-ig\hat{X}\otimes\hat{X}t}.$$

a. **(2 points)** Apply the time evolution operator to obtain combined state $|\psi(t)\rangle = \widehat{U}(t)|\psi_0\rangle$. Hint: You might want to use $|\psi_0\rangle$ in $\{|++\rangle, |+-\rangle, |-+\rangle, |--\rangle\}$ basis since these are eigenstates of $\hat{X} \otimes \hat{X}$.

b. **(4 points)** Find the density matrix of this state and expand it completely in $\{|++\rangle, |+-\rangle, |-+\rangle, |--\rangle\}$ basis. You should obtain $4 \times 4 = 16$ terms.

c. **(4 points)** Take the partial trace over the second qubit to obtain density matrix of the first:

$$\hat{\rho}_1(t) = \mathrm{Tr}_2[\hat{\rho}(t)] = \langle +_2|\hat{\rho}(t)|+_2\rangle + \langle -_2|\hat{\rho}(t)|-_2\rangle$$

Use $\frac{e^{i\theta}+e^{-i}}{2} = \cos\theta$ to simplify the expressions. Examples of partial trace:

$$\mathrm{Tr}_2[|+-\rangle\langle--|] = |+\rangle\langle-| \otimes [\langle+|(|-\rangle\langle-|)|+\rangle + \langle-|(|-\rangle\langle-|)|-\rangle]$$
$$= |+\rangle\langle-| \times [0+1] = |+\rangle\langle-|$$
$$\mathrm{Tr}_2[|+-\rangle\langle-+|] = |+\rangle\langle-| \otimes [\langle+|(|-\rangle\langle+|)|+\rangle + \langle-|(|-\rangle\langle+|)|-\rangle]$$
$$= |+\rangle\langle-| \times [0+0] = 0.$$

d. **(4 points)** Use the formulas for outer products of $|+\rangle, |-\rangle$ in terms of $|0\rangle, |1\rangle$

$$|+\rangle\langle+| = \frac{|0\rangle\langle0| + |0\rangle\langle1| + |1\rangle\langle0| + |1\rangle\langle1|}{2}, \qquad |+\rangle\langle-| = \frac{|0\rangle\langle0| - |0\rangle\langle1| + |1\rangle\langle0| - |1\rangle\langle1|}{2}$$

$$|-\rangle\langle+| = \frac{|0\rangle\langle0| + |0\rangle\langle1| - |1\rangle\langle0| - |1\rangle\langle1|}{2}, \qquad |-\rangle\langle-| = \frac{|0\rangle\langle0| - |0\rangle\langle1| - |1\rangle\langle0| + |1\rangle\langle1|}{2}$$

to obtain

$$\hat{\rho}_1(t) = \begin{bmatrix} \cos^2(gt) & 0 \\ 0 & \sin^2(gt) \end{bmatrix}.$$

This shows that qubit 1 which starts off in $|0\rangle$ oscillates between $|0\rangle$ and $|1\rangle$ due to interaction with qubit 2.

e. **(4 points)** Prove that this is generally a classical mixture and not a quantum superposition. Is it ever a pure state? If so, when?